Voice2Gov

A Citizen-Centric Civic Complaint Management Platform

ABSTRACT

Citizens often struggle to report local civic problems such as sewage overflow, damaged roads, or garbage accumulation. Once a complaint is made, there is rarely a clear way to know who received it, how urgent it is considered, or whether it has been resolved. Voice2Gov is a three-portal platform that creates a transparent, structured channel between citizens and government or civic organizations.

The citizen portal is a React Native mobile application. Citizens register and authenticate, then submit complaints with a category, description, photographic evidence from the camera or gallery, and confirmed location. Each complaint is routed to the appropriate organization and department, initially by category and later through location rules or AI-assisted classification. Eligible complaints appear in a community feed, where citizens can like or upvote them. A configurable priority score combines community engagement with severity, complaint age, and organization-set priority, so that likes influence visibility without being the sole factor.

The organization portal gives each organization an independent login and a tenant-isolated complaint queue. Staff can review evidence, assign complaints to departments or staff, post progress updates, and move complaints through a validated lifecycle: SUBMITTED, ONGOING, PROCESSING, and COMPLETED, with REJECTED, DUPLICATE, and CLOSED as alternate outcomes. The admin portal handles organizations, departments, categories, users, platform-wide analytics, and audit logs.

The backend is built with Python FastAPI and MongoDB. It exposes a versioned REST/JSON API documented with OpenAPI/Swagger, with JWT-based authentication and role-based access control for four roles: admin, organization admin, organization staff, and citizen. Security measures include Argon2 or bcrypt password hashing, Pydantic input validation, file upload validation, rate limiting, HTTPS, and audit logging. MongoDB uses geospatial (2dsphere) and compound indexes to support location-based queries and to prevent duplicate likes. Push notifications and an in-app notification center keep citizens informed at each key status change.

The system is planned in seven phases, from foundation through admin tooling to advanced features such as duplicate detection, heat maps, and Tamil/English multilingual support. Its guiding principle is that a citizen reports a problem once, the right organization receives it, the community helps prioritize it, and the citizen can follow its resolution from submission to completion.

Keywords: *civic complaints, e-governance, React Native, FastAPI, MongoDB, community prioritization, role-based access control*